

numpy: scalar, vector, matrix and tensor
 == dim 0 1 2 3 and above
 standard container images
 variables videos

slicing: extract the part of an array

$a = \text{np.array}([1, 2, 3, 4, 5])$

$\text{print}(a) \rightarrow [1 \ 2 \ 3 \ 4 \ 5]$
 0 1 2 3 4

$a[0] \rightarrow 1$

$a[::2]$

$\rightarrow 1$

$a[0:3] \rightarrow [1 \ 2 \ 3]$
 $\downarrow$
 $a[:3]$
 $\downarrow$
 default value = 0

$b = \text{np.arange}(1, 13).reshape((3, 4))$

	1	2	3	4
0	1	2	3	4
1	5	6	7	8
2	9	10	11	12

$b \begin{bmatrix} \cdot & \cdot \\ \cdot & \cdot \\ \text{rows} & \text{columns} \end{bmatrix}$

$b[1:3, 1:3] \rightarrow \begin{bmatrix} 6 & 7 \\ 10 & 11 \end{bmatrix}$

$b[1:, 1:3] \rightarrow \begin{bmatrix} 6 & 7 \\ 10 & 11 \end{bmatrix}$

$$b[1:3, 1:3] \rightarrow \begin{bmatrix} 6 & 7 \\ 10 & 11 \end{bmatrix}$$

	0	1	2	3
P	0	1	2	3
N	-1	-2	-3	-4
0	-1	1	2	3
1	-2	5	6	7
2	-1	9	10	11

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}, -1$$

$$\begin{bmatrix} 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 2 & 3 \\ 13 & 14 & 15 & 16 \\ 17 & 18 & 19 & 20 \\ 21 & 22 & 23 & 24 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 2 & 3 \\ 25 & 26 & 27 & 28 \\ 29 & 30 & 31 & 32 \\ 33 & 34 & 35 & 36 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

number of rows columns
or
matrix

$$C[2, 0:2, 1:3]$$

$$C[1, 1, :] \rightarrow C[1, 1]$$

$$C[2, 1:3, 1:3]$$